Generative-Model-Aided Grant-Free Random Access for GNSS-Independent MIMO-AFDM Satellite IoT Systems

I. Introduction

Low Earth orbit (LEO) satellites extend Internet-of-Things (IoT) connectivity to areas beyond the coverage of terrestrial networks. The sporadic transmission of short packets by a large population of terminals makes grant-free random access (GFRA) attractive for satellite IoT, since active terminals can transmit without first obtaining an uplink grant [1]. With limited pilot resources, the receiver must identify active terminals and estimate their channels from superimposed nonorthogonal pilot signals. This gives rise to the joint activity detection and channel estimation (JADCE) problem [2]. In LEO uplinks, the satellite motion and the different propagation distances across its coverage area introduce user-dependent delays and Doppler shifts, which further complicate pilot reception.

The Release 17 New Radio non-terrestrial network (NR-NTN) framework assumes terminals with global navigation satellite system (GNSS) capabilities. A terminal uses its GNSS position, satellite ephemeris, and common timing advance information to perform uplink timing and frequency pre-compensation [3, Secs. 16.14.1 and 16.14.2.2]. For low-cost, battery-powered IoT terminals, GNSS acquisition and the associated compensation procedures impose additional hardware and energy requirements [1]. This motivates GFRA schemes that operate without terminal GNSS assistance. In the setting considered here, a common reference delay is compensated, while the satellite receiver handles the remaining differential delays and Doppler shifts. These offsets can exceed a symbol duration and a subcarrier spacing, respectively, and therefore need to be retained in the signal model [1].

Waveforms designed for doubly dispersive channels have been investigated for satellite random access. Shen et al. proposed a spreading-based multi-frame orthogonal time-frequency space (OTFS) transmission scheme and a receiver for joint device identification, channel estimation, and symbol detection in the presence of large differential delays and Doppler shifts [1]. Affine frequency division multiplexing (AFDM) provides another approach through the discrete affine Fourier transform (DAFT), whose chirp parameters can be selected according to the delay and Doppler spreads [4]. For AFDM-based LEO GFRA, Ding et al. developed a graph neural network that refines sparse priors by learning correlations among delay–Doppler grid nodes for JADCE [5].

The choice of channel representation determines which dependencies a receiver can exploit. A discrete delay–Doppler representation describes the channel through path coefficients on a prescribed grid. Fractional delays and Doppler shifts spread the contribution of a physical path over multiple grid points, making the assumed support less accurate [1], [6]. An alternative is to estimate the AFDM effective channel directly [6]. In a multiantenna system, each terminal's effective channel comprises a collection of matrices that share path delays, Doppler shifts, and array responses. These physical parameters induce dependencies among the matrix entries and across receive antennas. Learning their joint distribution offers a way to use these dependencies in channel estimation without prescribing a discrete path support.

Score-based generative models learn a channel prior through the gradient of the log density of Gaussian-perturbed channel samples. Arvinte and Tamir applied such models to multiple-input multiple-output (MIMO) channel estimation using posterior sampling [7]. For LEO AFDM links, Huang et al. learned the score of the effective channel and performed inference using likelihood-annealed Langevin dynamics [6]. To incorporate generative priors into JADCE, Cai et al. combined channel denoising with message passing (MP) in a MIMO orthogonal frequency division multiplexing (OFDM) system [8]. Their receiver uses first- and second-order score networks to obtain channel moment estimates at each iteration, avoiding an online reverse diffusion sampling procedure.

In the GNSS-independent AFDM uplink, the observation model must also account for the unknown phase increments accumulated between pilot frames. These increments enter the sensing matrix, whereas the within-frame delay and Doppler effects enter the effective channel. Joint recovery of terminal activity and channels also requires evidence for the active and inactive hypotheses. We combine the learned channel moments with a density ratio for activity detection, allowing the MP receiver to use the joint distribution of complete multiantenna AFDM channel blocks for both channel estimation and activity detection.

In this paper, we investigate generative-model-aided GFRA for a GNSS-independent multiuser MIMO-AFDM satellite IoT system, where single-antenna terminals communicate with a LEO satellite equipped with a uniform planar array (UPA). Our main contributions are as follows.

1) We formulate a multi-frame AFDM uplink model with continuous path delays and Doppler shifts. A chirp-periodic prefix (CPP) accommodates the differential propagation delay, and the model separates inter-frame phase evolution

from the reference-frame effective channel. Stacking the received pilots yields a multiuser observation model for joint activity detection and full multiantenna channel estimation.

2) We develop a generative-model-aided MP receiver for JADCE. The linear estimator processes the multiuser pilot observations, while the nonlinear denoiser combines the learned channel prior with the terminal activity model. The two modules iteratively exchange extrinsic means and variances to update the channel estimates and activity probabilities. The inter-frame phase increments and noise variance are estimated from the received pilots before the MP iterations.
3) We incorporate a learned prior for the complete multiantenna AFDM effective channel into the nonlinear denoiser. Two score networks provide a channel mean estimate and a scalar error estimate. A separately trained classifier estimates the density ratio between active and inactive observations [9], which supplies the activity probabilities used in the channel estimates and outgoing messages. We specify the training objectives, message schedule, and computational complexity of the resulting receiver.

Numerical comparisons over 500 independent physical scenes show lower channel estimation error than the four evaluated baseline receivers over the tested signal-to-noise ratio (SNR) range of -4 to 8 dB, and lower activity detection error over -2 to 8 dB.

The remainder of this paper is organized as follows. Section II presents the system model and problem formulation. Section III develops the receiver, and Section IV describes the learning objectives. Section V summarizes the algorithm and analyzes its complexity. Numerical results and conclusions are presented in Sections VI and VII, respectively.
Notation: $\mathcal{CN}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ denotes a proper complex Gaussian law. $\|\cdot\|_F$ denotes the Euclidean norm of all matrix or tensor entries; $(\cdot)^{\mathsf{H}}$ and $\mathrm{tr}(\cdot)$ denote conjugate transpose and trace.

II. System Model

We consider a grant-free uplink from K single-antenna terminals to a low Earth orbit (LEO) satellite equipped with a uniform planar array (UPA). The array has $M = M_y M_z$ elements. An access block contains Q affine frequency division multiplexing (AFDM) pilot frames, each with N useful samples. Terminal activities, channel geometry, and path coefficients remain fixed within the block. The receiver knows the pilot sequences and the system parameters, whereas terminal activities, channel realizations, inter-frame phase increments, and receiver noise power are unknown.

Let $\alpha_k \in \{0, 1\}$ denote the activity state of terminal k. In the activity model, independent activations with probability p are conditioned on at least one transmission:

$$p_{\mathrm{gen}}(\boldsymbol{\alpha}) = \frac{\mathbf{1}\{\sum_{k=1}^{K} \alpha_k \geq 1\}}{Z} \prod_{k=1}^{K} p^{\alpha_k}(1-p)^{1-\alpha_k}, \qquad Z = 1 - (1-p)^K. \tag{1}$$

The resulting activity marginal is $\rho = p/Z$, which is supplied to the receiver.

A. Terrestrial-Satellite Link Model

Let $\mathbf{s}$ and $\mathbf{v}_{\mathrm{sat}}$ denote the satellite position and velocity, respectively. The position of terminal k is denoted by $\mathbf{g}_k$. With carrier frequency f_c and propagation speed c_0, define

$$\mathbf{u}_k = \frac{\mathbf{g}_k - \mathbf{s}}{\|\mathbf{g}_k - \mathbf{s}\|}, \qquad \nu_k = -\frac{f_c}{c_0} \mathbf{v}_{\mathrm{sat}}^{\mathsf{T}} \mathbf{u}_k. \tag{2}$$

We use the complex-envelope convention $s_{\mathrm{RF}}(t) = \Re\{x(t) e^{-j2\pi f_c t}\}$; thus, a path contributes $x(t - \tau(t)) e^{j2\pi f_c \tau(t)}$ up to its complex gain. Equation (2) consequently gives the Doppler time factor $e^{j2\pi\nu_k t}$. Envelope delays and array directions are held fixed over the block, while this carrier phase variation is retained.

After compensation of the common reference delay τ_c, the residual delay of path ℓ is given by

$$\tau_{k\ell} = \frac{\|\mathbf{g}_k - \mathbf{s}\|}{c_0} - \tau_c + \delta_\ell \tau_{\mathrm{spread},k}, \tag{3}$$

where τ_c is the propagation time to the closest ground point in the beam footprint, and $\delta_\ell \tau_{\mathrm{spread},k}$ is the excess multipath delay. We adopt a common spatial response and a common Doppler shift for the paths of each terminal [10, Sec. II-C] [6, Sec. II].

Let $\mu_{y,k} = \mathbf{e}_y^{\mathsf{T}} \mathbf{u}_k$ and $\mu_{z,k} = \mathbf{e}_z^{\mathsf{T}} \mathbf{u}_k$, where $\mathbf{e}_y$ and $\mathbf{e}_z$ are the array axes. For element position $\mathbf{s} - n_y d_y \mathbf{e}_y - n_z d_z \mathbf{e}_z$, the array response is

$$[\mathbf{a}_k]_m = \exp\left[j2\pi \left(\frac{d_y}{\lambda} n_y \mu_{y,k} + \frac{d_z}{\lambda} n_z \mu_{z,k} \right) \right], \quad m = n_z + M_z n_y, \quad \|\mathbf{a}_k\|^2 = M, \tag{4}$$

where $\lambda = c_0/f_c$, $0 \leq n_y < M_y$, and $0 \leq n_z < M_z$. The unknown array response is included in the channel to be estimated.

B. Transmission Scheme

Let $T_s = 1/(N\Delta f)$ be the sampling interval, where Δf denotes the subcarrier spacing. The discrete affine Fourier transform (DAFT) matrix is [4, Secs. II-B and III-A]

$$[\mathbf{F}_{\mathrm{A}}]_{rt} = \frac{1}{\sqrt{N}} e^{-j2\pi(c_2 r^2 + rt/N + c_1 t^2)}, \qquad r,t = 0,\dots,N-1. \tag{5}$$

For pilot vector $\mathbf{p}_{qk}$, the useful time samples are $\mathbf{s}_{qk} = \mathbf{F}_{\mathrm{A}}^{\mathsf{H}} \mathbf{p}_{qk}$. A chirp-periodic prefix (CPP) of length N_{CPP} extends these samples as

$$s_{qk}[t] = e^{j2\pi c_1 (t^2 - \langle t \rangle_N^2)} s_{qk}[\langle t \rangle_N], \qquad -N_{\mathrm{CPP}} \le t < 0, \tag{6}$$

where $\langle x \rangle_N = x - N\lfloor x/N \rfloor$. We choose $N_{\mathrm{CPP}} T_s \ge \max_{k,\ell} \tau_{k\ell}$ and $2Nc_1 \in \mathbb{N}_+$. The frame duration is $T_{\mathrm{frame}} = (N + N_{\mathrm{CPP}})T_s$.

To account for continuous delays, we use the AFDM waveform model in [6, Sec. II]. Let $w_\xi = \langle \xi \rangle_N$ and $\zeta_n(w) = \lfloor 2c_1 w + n/N \rfloor$. The basis is

$$\varphi_n(\xi) = \frac{1}{\sqrt{N}} e^{j2\pi[c_2 n^2 + c_1 \xi^2 + n w_\xi/N - \zeta_n(w_\xi) w_\xi]}. \tag{7}$$

It agrees with the inverse DAFT at integer useful-sample times and extends across prefix periods according to (6). For a delay $d = \tau/T_s$, set $w = \langle t - d \rangle_N$ and $\zeta_{tn} = \zeta_n(w)$. The delayed basis can be expressed as

$$[\mathbf{B}_d]_{tn} = \varphi_n(t-d) = \frac{1}{\sqrt{N}} e^{j2\pi[c_2 n^2 + c_1 (t-d)^2 + nw/N - \zeta_{tn} w]}. \tag{8}$$

The corresponding DAFT-domain path operator, for useful-window start index g, is given by

$$\mathbf{G}(d,\nu;g) = \mathbf{F}_{\mathrm{A}} \mathbf{D}_{\nu,g} \mathbf{B}_d, \qquad \mathbf{D}_{\nu,g} = \mathrm{diag}\{e^{j2\pi\nu T_s (g+t)}\}_{t=0}^{N-1}. \tag{9}$$

Expanding the path operator gives

$$\begin{aligned} G_{rn} = {} & \frac{e^{j2\pi[c_1 d^2 + c_2(n^2 - r^2) - nd/N + \nu T_s g]}}{N} \\ & \times \sum_{t=0}^{N-1} e^{j2\pi[(n-r)/N - 2c_1 d + \nu T_s] t} e^{j2\pi \zeta_{tn} d}. \end{aligned} \tag{10}$$

The last factor accounts for the fractional delay in the waveform. This expression corresponds to [6, eq. (5)] with normalized Doppler $k_{\nu,\mathrm{Huang}} = -\nu N T_s$ under our sign convention.

For integer d, (9) reduces to

$$\mathbf{G}(d,\nu;g) = \mathbf{F}_{\mathrm{A}} \mathbf{D}_{\nu,g} \mathbf{\Omega}_d \mathbf{\Pi}_{d \bmod N} \mathbf{F}_{\mathrm{A}}^{\mathsf{H}}, \tag{11}$$

$$[\mathbf{\Omega}_d]_{tt} = e^{j2\pi c_1 [(t-d)^2 - \langle t-d \rangle_N^2]}, \tag{12}$$

where $[\mathbf{\Pi}_l \mathbf{s}]_t = s[\langle t - l \rangle_N]$. The full delay enters the prefix phase, and its remainder determines the cyclic shift.

At the reference frame, the channel of terminal k is given by

$$\mathbf{G}_k = \sum_\ell g_{k\ell} \mathbf{G}(\tau_{k\ell}/T_s, \nu_k; N_{\mathrm{CPP}}), \qquad H_{kmrn} = [\mathbf{a}_k]_m [\mathbf{G}_k]_{rn}, \tag{13}$$

where $g_{k\ell}$ includes the path gain after large-scale power compensation. The tensor $H_k \in \mathbb{C}^{M \times N \times N}$ contains all antenna, DAFT output, and DAFT input coordinates. Reshaping it as a matrix in $\mathbb{C}^{M \times N^2}$ with one row per antenna yields a matrix of rank at most one under the common spatial response assumption.

Pilots are formed from selected rows and permuted columns of a KN-point discrete Fourier transform (DFT). With distinct row indices r_q, a column permutation c_{kn}, and independent uniform phases ϕ_{kn}, set

$$P_{qkn} = \sqrt{\frac{E_p}{N}} e^{-j2\pi r_q c_{kn}/(KN)} e^{j\phi_{kn}}, \qquad \sum_{n=0}^{N-1} |P_{qkn}|^2 = E_p. \tag{14}$$

The common terminal Doppler introduces the phase increment $\beta_k = \mathrm{wrap}(2\pi\nu_k T_{\mathrm{frame}})$ between consecutive frames, where wrap maps an angle to $[-\pi, \pi)$. The observation operator with these phase increments is

$$A_{qkn} = P_{qkn} e^{jq\beta_k}, \qquad q = 0,\dots,Q-1. \tag{15}$$

Define the effective channel $X_k = \alpha_k H_k$. After prefix removal and DAFT demodulation, the received signal can be expressed as

$$Y_{qmr} = \sum_{k=1}^{K} \sum_{n=0}^{N-1} A_{qkn} X_{kmrn} + W_{qmr}, \qquad W_{qmr} \sim \mathcal{CN}(0, \sigma^2), \tag{16}$$

with conditionally independent noise entries. Using the indexing

$$[\mathbf{Y}]_{q,mN+r} = Y_{qmr}, \quad [\mathbf{A}]_{q,(k-1)N+n} = A_{qkn}, \quad [\mathbf{X}]_{(k-1)N+n,mN+r} = X_{kmrn}, \tag{17}$$

the multiple measurement vector (MMV) model can be written as

$$\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{W}, \quad \mathbf{Y} \in \mathbb{C}^{Q\times MN}, \quad \mathbf{A} \in \mathbb{C}^{Q\times KN}, \quad \mathbf{X} \in \mathbb{C}^{KN\times MN}. \tag{18}$$

All MN measurement vectors share the terminal activity pattern. The receiver takes the observation tensor, the original pilots P, and the activity marginal ρ as inputs. It estimates $\widehat{\boldsymbol{\beta}}$ and $\widehat{\sigma}^2$ from these observations and constructs $\widehat{\mathbf{A}} = \mathbf{A}(\widehat{\boldsymbol{\beta}})$. The physical operator $\mathbf{A} = \mathbf{A}(\boldsymbol{\beta})$ and the noise variance σ^2 in (18) describe data generation. For simulations, the signal-to-noise ratio (SNR) is defined using the clean superposition $\mathbf{Y}_0 = \mathbf{A}\mathbf{X}$:

$$\sigma^2 = \frac{\|\mathbf{Y}_0\|_F^2}{QMN} 10^{-\mathrm{SNR}_{\mathrm{dB}}/10}. \tag{19}$$

The received signal power includes cross terms between terminals.

C. Problem Formulation

Let $D = MN^2$ and let q_{tr} be the distribution of normalized channels generated from the physical model before applying activity masks. We normalize channels and observations by a common scale s, the root mean square of the training channels: $(\mathbf{Y}, H_k, X_k, \mathbf{W})$ are divided by s, and both σ^2 and $\widehat{\sigma}^2$ by s^2. The pilot operators are unchanged. The same symbols denote the normalized quantities below; the reported channel estimates are rescaled by s.

For joint activity detection and channel estimation (JADCE), we use the factorized prior

$$q(\{dH_k\}, \boldsymbol{\alpha}) = \prod_{k=1}^{K} q_{\mathrm{tr}}(dH_k)\rho^{\alpha_k}(1-\rho)^{1-\alpha_k}. \tag{20}$$

We approximate the joint prior by the product of its channel and activity marginals. With the estimated phase and noise variance held fixed, the approximate posterior distribution can be expressed as

$$\begin{aligned} &q(d\mathbf{X}, \{dH_k\}, \boldsymbol{\alpha} \mid \mathbf{Y}, \widehat{\mathbf{A}}, \widehat{\sigma}^2) \\ &\quad \propto e^{-\left\|\mathbf{Y}-\widehat{\mathbf{A}}\mathbf{X}\right\|_F^2/\widehat{\sigma}^2} \prod_{k=1}^{K} \delta_{\alpha_k H_k}(dX_k) q_{\mathrm{tr}}(dH_k)\rho^{\alpha_k}(1-\rho)^{1-\alpha_k}, \end{aligned} \tag{21}$$

where $\delta_{\alpha_k H_k}$ enforces $X_k = \alpha_k H_k$. Under this posterior, the minimum mean-squared error (MMSE) target and the activity decision are given by

$$\begin{aligned} X_k^{\mathrm{MMSE}} &= \mathbb{E}_q[X_k \mid \mathbf{Y}, \widehat{\mathbf{A}}, \widehat{\sigma}^2], \\ p_k^{\mathrm{Bayes}} &= q(\alpha_k = 1 \mid \mathbf{Y}, \widehat{\mathbf{A}}, \widehat{\sigma}^2), \qquad \alpha_k^{\mathrm{Bayes}} = \mathbf{1}\{p_k^{\mathrm{Bayes}} \geq \tfrac{1}{2}\}. \end{aligned} \tag{22}$$

The threshold corresponds to equal costs for a missed detection and a false alarm. The receiver approximates these targets by iterating between linear estimation and channel and activity denoising, returning $\widehat{X}_k$, p_k, and $\widehat{\alpha}_k = \mathbf{1}\{p_k \geq 1/2\}$.

III. Bayesian Receiver Design

We first estimate the phase increments and noise variance from the received pilots. With these estimates fixed, the receiver uses message passing (MP) between a linear estimator and a nonlinear denoiser. The two modules iteratively exchange extrinsic messages [11]. These messages are obtained by Gaussian projection and message division [12]. The nonlinear denoiser uses the learned channel moments and the density ratio for activity detection. The corresponding learning objectives are presented in Section IV.

A. Parameter Estimation

1) Inter-frame phase estimation: Stack the received samples as $[\mathbf{Y}_{\mathrm{s}}]_{qN+r,m} = Y_{qmr}$, so that $\mathbf{Y}_{\mathrm{s}} \in \mathbb{C}^{QN\times M}$. To estimate the phase increments, we approximate the contribution of each candidate terminal by one dominant path with an unconstrained complex gain at each antenna. Absorbing the common scalar phase into the gain, we define the path matrix as

$$\mathbf{G}_0(d,\nu) = e^{-j2\pi c_1 d^2}\mathbf{G}(d,\nu;0), \qquad [\mathbf{h}_k(d,\nu)]_{qN+r} = e^{j2\pi\nu q T_{\mathrm{frame}}} \sum_{n=0}^{N-1} [\mathbf{G}_0(d,\nu)]_{rn} P_{qkn}. \tag{23}$$

The delay search is restricted to $d \in [0, N)$ and the Doppler search to $|\nu| \le \nu_{\max}$. Under $2Nc_1 \in \mathbb{N}_+$, shifting the delay by an integer multiple of N changes the path operator by a scalar phase, which the fitted gains absorb. With $\mathbf{h}_k = \mathbf{h}_k(d_k, \nu_k)$ and gain vector $\mathbf{b}_k \in \mathbb{C}^M$, the fitting objective is

$$\min_{\{d_k, \nu_k, \mathbf{b}_k\}} \left\| \mathbf{Y}_{\mathrm{s}} - \sum_{k=1}^{K} \mathbf{h}_k \mathbf{b}_k^{\mathsf{T}} \right\|_F^2 . \tag{24}$$

When updating terminal k, we fix the other contributions and define $\mathbf{Z}_k = \mathbf{Y}_{\mathrm{s}} - \sum_{j \neq k} \mathbf{h}_j \mathbf{b}_j^{\mathsf{T}}$. For a given (d, ν), the least-squares gain and the resulting objective are given by

$$\begin{aligned} \mathbf{b}_k^{\mathsf{T}}(d, \nu) &= \frac{\mathbf{h}_k(d, \nu)^{\mathsf{H}} \mathbf{Z}_k}{\|\mathbf{h}_k(d, \nu)\|^2}, \\ J_k(d, \nu) &= \|\mathbf{Z}_k\|_F^2 - \frac{\|\mathbf{h}_k(d, \nu)^{\mathsf{H}} \mathbf{Z}_k\|^2}{\|\mathbf{h}_k(d, \nu)\|^2}. \end{aligned} \tag{25}$$

The gains are eliminated from the separable nonlinear least-squares problem by variable projection [13, Secs. 1 and 2]. Coarse correlations over delay, within-frame Doppler, and frame phase provide initial candidates. A phase candidate β corresponds to Doppler aliases

$$\nu_j = \frac{\beta + 2\pi j}{2\pi T_{\mathrm{frame}}}, \qquad j \in \mathbb{Z}, \quad |\nu_j| \le \nu_{\max}. \tag{26}$$

Candidates are ranked using the pilot response vector in (23), which enforces the same Doppler within and across frames. The selected candidates undergo bounded local refinement by L-BFGS-B [14], with delay restricted to one differentiable branch of the wrapped waveform. Each coordinate update is accepted when it does not increase the residual energy, up to numerical tolerance. After each sweep, all antenna gains are refitted jointly by least squares.

The receiver uses a fixed number of complete sweeps, retaining the previous position as an additional refinement starting point after the first sweep. It sets $\widehat{\beta}_k = \mathrm{wrap}(2\pi \widehat{\nu}_k T_{\mathrm{frame}})$ and constructs $\widehat{\mathbf{A}}$ by (15). This local fit provides phase estimates for all candidate terminals. The nonlinear denoiser then estimates the full channel tensor in (13).

2) Noise variance estimation: Let $L = QN > M \ge 2$ and form the spatial sample covariance

$$\widehat{\mathbf{R}}_y = \frac{1}{L} \sum_{q=0}^{Q-1} \sum_{r=0}^{N-1} \mathbf{y}_{qr} \mathbf{y}_{qr}^{\mathsf{H}}, \qquad \mathbf{y}_{qr} = (Y_{q0r}, \ldots, Y_{q,M-1,r})^{\mathsf{T}}. \tag{27}$$

Write its eigenvalues in descending order as $\lambda_1 \ge \cdots \ge \lambda_M$; values below $\epsilon_{\mathrm{mach}} \lambda_1$ are floored at that level before taking logarithms. For a candidate spatial signal rank j, define

$$\bar{\lambda}_j = \frac{1}{M - j} \sum_{i=j+1}^{M} \lambda_i, \qquad \log g_j = \frac{1}{M - j} \sum_{i=j+1}^{M} \log \lambda_i. \tag{28}$$

The Wax–Kailath minimum description length (MDL) criterion, as reviewed in [15, Sec. II-A, eqs. (5)–(7)], gives

$$\begin{aligned} \widehat{r}_{\mathrm{s}} &= \arg \min_{0 \le j < M} \left\{ L(M - j)(\log \bar{\lambda}_j - \log g_j) + \frac{j(2M - j)}{2} \log L \right\}, \\ \widehat{\sigma}^2 &= \bar{\lambda}_{\widehat{r}_{\mathrm{s}}}. \end{aligned} \tag{29}$$

Under a lower-rank spatial signal plus white-noise model, the criterion selects the spatial signal rank and estimates the noise variance from the remaining eigenvalues. Both $\widehat{\sigma}^2$ and $\widehat{\mathbf{A}}$ remain fixed throughout the MP iterations.

B. Linear Estimation

Let the incoming Gaussian message have mean $\mathbf{X}^{\mathrm{pri}} \in \mathbb{C}^{KN \times MN}$ and covariance $\gamma \mathbf{I}_{KN}$ per measurement vector, where $\gamma > 0$. For $\widehat{\sigma}^2 > 0$, the posterior mean and covariance under the estimated operator in (21) are given by

$$\mathbf{X}_{\mathrm{L}} = \mathbf{X}^{\mathrm{pri}} + \gamma \widehat{\mathbf{A}}^{\mathsf{H}} (\widehat{\sigma}^2 \mathbf{I}_Q + \gamma \widehat{\mathbf{A}} \widehat{\mathbf{A}}^{\mathsf{H}})^{-1} (\mathbf{Y} - \widehat{\mathbf{A}} \mathbf{X}^{\mathrm{pri}}), \tag{30}$$

$$\mathbf{\Sigma}_{\mathrm{L}} = (\gamma^{-1} \mathbf{I}_{KN} + \widehat{\sigma}^{-2} \widehat{\mathbf{A}}^{\mathsf{H}} \widehat{\mathbf{A}})^{-1}, \qquad t_{\mathrm{L}} = \frac{\operatorname{tr} \mathbf{\Sigma}_{\mathrm{L}}}{KN}. \tag{31}$$

We approximate $\mathbf{\Sigma}_{\mathrm{L}}$ by $t_{\mathrm{L}} \mathbf{I}_{KN}$, where t_{L} is its normalized trace. Dividing the resulting Gaussian approximation by the incoming Gaussian message gives

$$v_{\mathrm{L}} = (t_{\mathrm{L}}^{-1} - \gamma^{-1})^{-1}, \qquad \mathbf{R}_{\mathrm{L}} = v_{\mathrm{L}} \left(\frac{\mathbf{X}_{\mathrm{L}}}{t_{\mathrm{L}}} - \frac{\mathbf{X}^{\mathrm{pri}}}{\gamma} \right). \tag{32}$$

For $\widehat{\mathbf{A}} \neq \mathbf{0}$, the eigenvalues in (31) give $0 < t_{\mathrm{L}} < \gamma$, so v_{L} is positive.

A thin singular value decomposition (SVD) $\widehat{\mathbf{A}} = \mathbf{U}\,\mathrm{diag}(s_i)\mathbf{V}^{\mathsf{H}}$ gives

$$f_{\mathrm{L}} = \frac{1}{KN}\sum_i \frac{\gamma s_i^2}{\widehat{\sigma}^2 + \gamma s_i^2}, \qquad t_{\mathrm{L}} = \gamma(1 - f_{\mathrm{L}}), \qquad v_{\mathrm{L}} = \frac{t_{\mathrm{L}}}{f_{\mathrm{L}}}. \tag{33}$$

Then $\mathbf{R}_{\mathrm{L}} = \mathbf{X}^{\mathrm{pri}} + (\mathbf{X}_{\mathrm{L}} - \mathbf{X}^{\mathrm{pri}})/f_{\mathrm{L}}$. The SVD is reused throughout the block. After reshaping by (17) and applying the damping in Section V-B, the nonlinear estimator receives (R_k, v) for each terminal.

C. Channel and Activity Denoising

For channel and activity denoising, we use the following additive white Gaussian noise (AWGN) observation model [8, Secs. III-B and V-B1]:

$$R_k = \alpha_k H_k + W_k, \qquad H_k \sim q_{\mathrm{tr}}, \qquad \mathrm{vec}(W_k) \sim \mathcal{CN}(\mathbf{0}, v\mathbf{I}_D). \tag{34}$$

This model generates the training samples and approximates the iterative receiver inputs. Before damping, the linear extrinsic mean satisfies

$$\begin{aligned} \mathbf{R}_{\mathrm{L}} - \mathbf{X} &= (\mathbf{L}\widehat{\mathbf{A}} - \mathbf{I})(\mathbf{X} - \mathbf{X}^{\mathrm{pri}}) \\ &\quad + \mathbf{L}(\mathbf{A} - \widehat{\mathbf{A}})\mathbf{X} + \mathbf{L}\mathbf{W}, \\ \mathbf{L} &= \frac{\gamma}{f_{\mathrm{L}}}\widehat{\mathbf{A}}^{\mathsf{H}}(\widehat{\sigma}^2\mathbf{I}_Q + \gamma\widehat{\mathbf{A}}\widehat{\mathbf{A}}^{\mathsf{H}})^{-1}. \end{aligned} \tag{35}$$

The three terms represent feedback error, error in the observation matrix due to phase estimation, and filtered thermal noise. Equation (34) approximates their sum by isotropic Gaussian noise independent of the channel.

The input densities for active and inactive terminals are given by

$$\begin{aligned} q_{1,v}(R) &= \int \mathcal{CN}(\mathrm{vec}(R); \mathrm{vec}(H), v\mathbf{I}_D) q_{\mathrm{tr}}(dH), \\ q_{0,v}(R) &= \mathcal{CN}(\mathrm{vec}(R); \mathbf{0}, v\mathbf{I}_D). \end{aligned} \tag{36}$$

Bayes' rule gives the local joint posterior

$$q(dH, a \mid R, v) = \frac{\rho^a (1-\rho)^{1-a} \mathcal{CN}(\mathrm{vec}(R); a\,\mathrm{vec}(H), v\mathbf{I}_D) q_{\mathrm{tr}}(dH)}{(1-\rho)q_{0,v}(R) + \rho q_{1,v}(R)}. \tag{37}$$

Hence the local activity probability is

$$p^\star(R, v) = \frac{\rho q_{1,v}(R)}{(1-\rho)q_{0,v}(R) + \rho q_{1,v}(R)}. \tag{38}$$

Let $m(R, v) = \mathbb{E}[H \mid R, v, \alpha = 1]$ and $\mathbf{\Sigma}_H(R, v) = \mathrm{Cov}(\mathrm{vec}(H) \mid R, v, \alpha = 1)$. Using the laws of total expectation and covariance, we have

$$\begin{aligned} \mathbb{E}[X \mid R, v] &= p^\star m, \\ \mathrm{Cov}(\mathrm{vec}(X) \mid R, v) &= p^\star \mathbf{\Sigma}_H + p^\star(1 - p^\star)\,\mathrm{vec}(m)\,\mathrm{vec}(m)^{\mathsf{H}}. \end{aligned} \tag{39}$$

This activity-weighted conditional mean is the standard Bayesian denoising construction [2, Sec. III-C]. Its evaluation requires channel moments and the density ratio $q_{1,v}/q_{0,v}$, supplied by the channel and activity estimators, respectively.

We approximate these quantities using a channel mean $F_k = F_\theta(R_k, v)$, a scalar conditional mean-squared error $\widehat{\tau}_k = \widehat{\tau}_\psi(R_k, v)$, and an activity logit $z_\phi(R_k, v, F_k)$. We average the conditional mean-squared error estimates over the K terminals and apply a positive lower bound:

$$\tau_{\mathrm{raw}} = \frac{1}{K}\sum_{k=1}^{K} \widehat{\tau}_k, \qquad \tau = \max\{\tau_{\mathrm{raw}}, \tau_{\min}\}, \tag{40}$$

where $\tau_{\min} > 0$. The resulting scalar τ is used as the variance per coordinate in the Gaussian approximation for each terminal. The training targets are specified in Section IV; Appendix A derives this averaging operation by fitting the Gaussian approximations.

For each terminal, retain a Gaussian approximation to the channel-prior factor, called a site [16, Sec. 2.1], of the form $\mathcal{CN}(\mathrm{vec}(H_k); \mathrm{vec}(E_k), c_k\mathbf{I}_D)$. For an input variance within the training range and $0 < \tau < v$, dividing the Gaussian approximation to the conditional posterior by the input likelihood gives

$$c_k = (\tau^{-1} - v^{-1})^{-1}, \qquad E_k = c_k\left(\frac{F_k}{\tau} - \frac{R_k}{v}\right). \tag{41}$$

This condition ensures positive site precision; otherwise, the previous site is retained as described in Section V-A. Combining the Gaussian site with the current input, the conditional variance and mean are respectively given by

$$t_k = \frac{c_k v}{c_k + v}, \qquad \mu_k = \frac{c_k R_k + v E_k}{c_k + v}. \tag{42}$$

For an accepted update, $\mu_k = F_k$ and $t_k = \tau$. If the update is rejected, these quantities instead combine the retained site with the current input.

Within the training variance range, the activity probability is

$$\ell_k = z_\phi(R_k, v, F_k) - \operatorname{logit} \pi_{\mathrm{tr}}, \qquad p_k = \operatorname{sigmoid}(\operatorname{logit} \rho + \ell_k), \tag{43}$$

where π_{tr} is the training class prior, $\operatorname{logit} a = \log[a/(1-a)]$, and $\operatorname{sigmoid} b = (1 + e^{-b})^{-1}$. Section IV-C derives this prior correction. The resulting local posterior approximation can be expressed as

$$\begin{aligned} \widetilde{q}_k(dX_k) &= (1 - p_k)\delta_0(dX_k) + p_k \mathcal{CN}(\mathrm{vec}(X_k); \mathrm{vec}(\mu_k), t_k \mathbf{I}_D)\, dX_k, \\ \widehat{X}_k &= p_k \mu_k. \end{aligned} \tag{44}$$

Let $\mathbf{U}_k \in \mathbb{C}^{M \times N^2}$ be obtained by reshaping μ_k with one row per antenna. Averaging the covariance across its N^2 columns gives

$$\mathbf{C}_{X,k} = p_k t_k \mathbf{I}_M + \frac{p_k(1 - p_k)}{N^2} \mathbf{U}_k \mathbf{U}_k^{\mathsf{H}}. \tag{45}$$

The first term represents conditional channel uncertainty; the second accounts for uncertainty in terminal activity.

D. Extrinsic Message Update

Averaging the traces of $\mathbf{C}_{X,k}$ over terminals and dividing by the number of antennas gives the scalar variance used for feedback:

$$\bar{t}_X = \frac{1}{KM} \sum_k \operatorname{tr} \mathbf{C}_{X,k} = \frac{1}{K} \sum_k \left[p_k t_k + p_k(1 - p_k) \frac{\|\mu_k\|_F^2}{D} \right]. \tag{46}$$

Let $b = \mathrm{clip}(\bar{t}_X / v, \varepsilon, 1 - \varepsilon)$, with $0 < \varepsilon < 1/2$. Dividing the Gaussian with mean $\widehat{X}_k$ and variance bv by the incoming Gaussian message gives

$$\gamma_{\mathrm{out}} = \frac{vb}{1 - b}, \qquad X_{k,\mathrm{out}} = \frac{\widehat{X}_k - bR_k}{1 - b}. \tag{47}$$

Clipping bounds the variance ratio and keeps the outgoing precision positive. When clipping is inactive, $bv = \bar{t}_X$. The damped outgoing parameters become the next linear prior.

IV. Learning Channel and Activity Denoisers with Score-Based Generative Priors

The channel prior is learned through the scores of its Gaussian-perturbed distributions [7], [17]. Following [8, Sec. IV], we train two score networks to estimate the conditional channel moments. We also train a classifier to estimate the density ratio for activity detection in (38). The training samples are generated according to the Gaussian observation model in (34).

A. First-Order Denoising Score Matching

For an active terminal, let $\mathbf{u} = (\Re \operatorname{vec}(R), \Im \operatorname{vec}(R)) \in \mathbb{R}^{2D}$ and define $\mathbf{h}_{\mathbb{R}}$ similarly. The real noise variance is $v_{\mathbb{R}} = v/2$, so that $\mathbf{u} = \mathbf{h}_{\mathbb{R}} + \sqrt{v_{\mathbb{R}}}\boldsymbol{\epsilon}$, with $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_{2D})$. Let $q_v^{\mathbb{R}}$ denote the smooth positive density obtained by Gaussian perturbation. Assuming finite moments, the conditional mean is given by Tweedie's formula [18]

$$\mathbf{m}_{\mathbb{R}}(\mathbf{u}, v) = \mathbf{u} + v_{\mathbb{R}} \nabla_{\mathbf{u}} \log q_v^{\mathbb{R}}(\mathbf{u}). \tag{48}$$

We parameterize the score by $S_\theta(\mathbf{u}, v) \in \mathbb{R}^{2D}$ and minimize the denoising score matching objective [17], [19]

$$\mathcal{L}_1(\theta) = \mathbb{E}\left[\frac{1}{2D} \left\| \sqrt{v_{\mathbb{R}}} S_\theta(\mathbf{u}, v) + \boldsymbol{\epsilon} \right\|^2 \right]. \tag{49}$$

At each positive noise level, its unrestricted population minimizer is $\nabla_{\mathbf{u}} \log q_v^{\mathbb{R}}(\mathbf{u})$. The corresponding complex channel estimate is given by

$$F_\theta(R, v) = R + \frac{v}{2}\big(S_{\theta,\Re}(R, v) + j S_{\theta,\Im}(R, v)\big), \tag{50}$$

where the two score components are reshaped to $M \times N \times N$. The channel mean estimate in (41) is thus obtained by a single network evaluation.

B. Second-Order Denoising Score Matching

Using the second-order form of Tweedie's formula, the conditional covariance can be expressed as

$$\mathbf{C}_{\mathbb{R}}(\mathbf{u}, v) = v_{\mathbb{R}}\mathbf{I}_{2D} + v_{\mathbb{R}}^2 \nabla_{\mathbf{u}}^2 \log q_v^{\mathbb{R}}(\mathbf{u}). \tag{51}$$

To estimate the scalar error required by the receiver, we adopt the trace objective for second-order denoising score matching [20, Corollary 4.2]. For a network $T_\psi(\mathbf{u}, v) \in \mathbb{R}^{2D}$, define

$$a_\psi(R, v) = 1 + \frac{v_{\mathbb{R}}}{2D} \sum_{i=1}^{2D} T_{\psi,i}(\mathbf{u}, v), \qquad \widehat{\tau}_\psi(R, v) = v\, a_\psi(R, v). \tag{52}$$

Fix S_θ and let $\mathbf{b} = \sqrt{v_{\mathbb{R}}} S_\theta + \boldsymbol{\epsilon}$. We train the second network by minimizing

$$\begin{aligned} \mathcal{L}_2(\psi) &= \mathbb{E}\left[\left\{\frac{1}{2D}\sum_{i=1}^{2D}(v_{\mathbb{R}} T_{\psi,i} - b_i^2 + 1)\right\}^2\right] \\ &= \mathbb{E}\left[\left\{a_\psi(R, v) - \frac{\|F_\theta(R, v) - H\|_F^2}{Dv}\right\}^2\right]. \end{aligned} \tag{53}$$

The average over coordinates is taken before squaring. The scalar output in (52) estimates the conditional mean-squared error of the channel estimator. Averaging these estimates over terminals and applying the lower bound in (40) gives τ, the variance per coordinate used in the Gaussian approximation for each terminal in (41). Appendix A derives the learning target and the Gaussian fitting objective that yields this average.

C. Density Ratio Estimation

The activity posterior in (38) depends on $q_{1,v}/q_{0,v}$. We learn this ratio from active-channel and noise-only samples using the relationship between density ratios and class probabilities [9, Secs. 3 and 4.2]. Both classes use the same sampling law for v, and the training class prior is $\pi_{\mathrm{tr}} = 1/2$. For label $a \in \{0, 1\}$, we minimize the binary cross-entropy (BCE) loss

$$\mathcal{L}_{\mathrm{act}}(\phi) = \mathbb{E}[\log(1 + e^{z_\phi}) - a z_\phi], \qquad z_\phi = z_\phi(R, v, F_\theta(R, v)). \tag{54}$$

The mean network remains fixed during this training.

Let $\eta_{\mathrm{tr}}(R, v) = \Pr_{\mathrm{tr}}(a = 1 \mid R, v)$. The derivative of the conditional BCE risk with respect to the logit is $\mathrm{sigmoid}(z) - \eta_{\mathrm{tr}}$, with strictly positive second derivative. Applying Bayes' rule, the optimal logit is given by

$$z^\star(R, v) = \mathrm{logit}\, \pi_{\mathrm{tr}} + \log \frac{q_{1,v}(R)}{q_{0,v}(R)}. \tag{55}$$

The sampling density of v is identical for both classes and cancels in these odds. Since the classifier observes R, the deterministic feature $F_\theta(R, v)$ does not change the optimal logit. Subtracting $\mathrm{logit}\, \pi_{\mathrm{tr}}$ and adding $\mathrm{logit}\, \rho$ yields the receiver probability in (43). For fixed class-conditional input distributions, the same classifier thus accommodates a different access marginal through the prior-odds correction.

D. Offline Training

Training channels are drawn before activity masking, and training and validation sets use disjoint physical scenes. For the channel networks, draw $H \sim q_{\mathrm{tr}}$, sample $\log v$ uniformly over $[\log v_{\min}, \log v_{\max}]$, and generate $R = H + W$ with $\mathrm{vec}(W) \sim \mathcal{CN}(\mathbf{0}, v\mathbf{I}_D)$. The real and imaginary antenna maps form $2M$ input channels over the $N \times N$ DAFT coordinates; the networks are conditioned on the input noise variance. The output of S_θ has the same shape, while T_ψ is reduced to the scalar in (52).

For classifier training, active and inactive samples at a sampled variance are $R_1 = H + W_1$ and $R_0 = W_0$, with independent Gaussian noises. The classifier receives the real and imaginary maps of R and F_θ, scaled by $\sqrt{2/(1+v)}$, together with

$$\begin{aligned} \Big(&\log v/c_v,\ \log(1 + e_R),\ \log(1 + e_F), \\ &\mathrm{sgn}(c_{RF}) \log(1 + |c_{RF}|),\ \log(1 + e_{\mathrm{res}})\Big), \end{aligned} \tag{56}$$

where $c_v > 0$ is a fixed conditioning scale, $e_R = \|R\|_F^2/(Dv)$, $e_F = \|F_\theta\|_F^2/(Dv)$, $c_{RF} = \Re\langle R, F_\theta\rangle_F/(Dv)$, and $e_{\mathrm{res}} = \|R - F_\theta\|_F^2/(Dv)$. The feature maps have $4M$ channels, and the classifier returns one logit per terminal.

We first train S_θ using (49). We then freeze it and train T_ψ and z_ϕ using (53) and (54), respectively. All networks remain fixed during online inference. They are conditioned on the denoiser input variance v, while the linear module

uses the physical noise estimate $\widehat{\sigma}^2$. Each network operates on a terminal's channel input at fixed dimensions M and N, allowing the same networks to be evaluated for different K and Q. Dataset sizes, network dimensions, and optimization settings are specified in the simulation setup.

V. Overall Algorithm

The receiver combines the estimators and message updates in Section III with the trained networks in Section IV. The phase increments and noise variance are estimated once per access block, followed by MP iterations between linear estimation and channel and activity denoising.

A. Message Scheduling

From the original received pilots, estimate $\widehat{\sigma}^2$ by (29), fit the inter-frame phase increments using (25), and form $\widehat{\mathbf{A}}$. Normalize the observation and estimated noise variance by s and s^2, respectively. Initialize $\mathbf{X}^{\mathrm{pri},(0)} = \mathbf{0}$ and $\gamma^{(0)} = \rho\widehat{P}_H$, where $\widehat{P}_H$ is the training average of $\|H\|_F^2/D$ in normalized units. The initial sites are $E_k^{(0)} = 0$ and $c_k^{(0)} = 1$ in these units. An activity history $w_k^{(0)} = \rho$ is maintained for the stopping check in Section V-B.

At each iteration, the linear estimator produces a Gaussian extrinsic message using (30)–(33). For $v \in [v_{\min}, v_{\max}]$, each trained network is evaluated once per terminal. The channel error estimates are averaged according to (40), and the sites are replaced using (41) if $\tau < v$; otherwise, the previous sites are retained. The classifier uses the current (R_k, v, F_k) in either case.

For v outside the training range, the sites are retained and the activity probability is obtained from their Gaussian densities:

$$\begin{aligned} \ell_k^{\mathrm{G}} &= \frac{\|R_k\|_F^2}{v} - \frac{\|R_k - E_k\|_F^2}{c_k + v} - D\log\left(1 + \frac{c_k}{v}\right), \\ p_k &= \mathrm{sigmoid}(\mathrm{logit}\,\rho + \ell_k^{\mathrm{G}}). \end{aligned} \tag{57}$$

The active density is $\mathcal{CN}(\mathrm{vec}(R_k); \mathrm{vec}(E_k), (c_k+v)\mathbf{I}_D)$, obtained by integrating the Gaussian site against the Gaussian likelihood. Its ratio to $q_{0,v}$ gives the approximate activity evidence in (57).

The joint estimate and feedback message are then calculated from the conditional moments and activity probabilities. The complete procedure is summarized in Algorithm 1.

Algorithm 1: Joint activity detection and channel estimation

Input: received pilots $\mathbf{Y}$, original pilot tensor P, and activity marginal ρ; trained S_θ, T_ψ, z_ϕ and fixed receiver settings.

Output: channel estimates, activity probabilities and decisions.

1) Estimate $\widehat{\sigma}^2$ using (29). Fit $\widehat{\boldsymbol{\beta}}$ by the coordinate sweeps and joint gain refits in Section III-A; construct $\widehat{\mathbf{A}}$.
2) Normalize $\mathbf{Y}$ and $\widehat{\sigma}^2$ by s and s^2. Initialize the prior, sites, activity history, and stability counters. Compute the thin SVD of $\widehat{\mathbf{A}}$ and hold both estimated observation parameters fixed.
3) For $i = 1, \ldots, I_{\max}$, compute the linear extrinsic message using (30)–(33) and damp it by (58).
4) If v is within the training range, evaluate F_k, $\widehat{\tau}_k$, and $z_\phi(R_k, v, F_k)$; average the conditional mean-squared error estimates over terminals and apply the lower bound in (40). Accept the site update (41) when $\tau < v$ and otherwise retain the sites. Compute p_k by (43).
5) If v is outside the training range, retain the sites and compute p_k by (57).
6) Compute μ_k, t_k by (42) and $\widehat{X}_k, \mathbf{C}_{X,k}$ by (44)–(45).
7) Form the outgoing message using (46)–(47) and apply (59).
8) Update the stability counters and activity history. Stop when the criterion in Section V-B is satisfied.
   end for
9) Return $s\widehat{X}_k$, $s\mu_k$, p_k, $\widehat{\alpha}_k = \mathbf{1}\{p_k \geq 1/2\}$, and the stopping status. Rescale covariance outputs by s^2.

B. Damping and Stopping Criteria

We apply damping to the means and variances of both the denoiser input and the feedback message. With new-message weight $0 < \eta \leq 1$, the input update for $i \geq 2$ is

$$\begin{aligned} R_k^{(i)} &= (1-\eta)R_k^{(i-1)} + \eta R_{\mathrm{L},k}^{(i)}, \\ v^{(i)} &= (1-\eta)v^{(i-1)} + \eta v_{\mathrm{L}}^{(i)}. \end{aligned} \tag{58}$$

Similarly, the feedback parameters are updated as

$$\begin{aligned} X_k^{\mathrm{pri},(i)} &= (1-\eta)X_k^{\mathrm{pri},(i-1)} + \eta X_{k,\mathrm{out}}^{(i)}, \\ \gamma^{(i)} &= (1-\eta)\gamma^{(i-1)} + \eta\gamma_{\mathrm{out}}^{(i)}. \end{aligned} \tag{59}$$

The first updates are used without damping.

The stopping rule tracks the estimate, activity probabilities, and input variance. For $i \geq 2$, define

$$\begin{aligned} d_X^{(i)} &= \frac{\left\| \widehat{X}^{(i)} - \widehat{X}^{(i-1)} \right\|_F}{\max\{\left\| \widehat{X}^{(i)} \right\|_F, \left\| \widehat{X}^{(i-1)} \right\|_F, \epsilon_{\mathrm{num}}\}}, \\ d_p^{(i)} &= \max_k |p_k^{(i)} - p_k^{(i-1)}|, \\ d_v^{(i)} &= \frac{|v^{(i)} - v^{(i-1)}|}{\max\{v^{(i)}, v^{(i-1)}\}}, \end{aligned} \tag{60}$$

where $\epsilon_{\mathrm{num}} > 0$ prevents division by zero. The estimate-stability counter increases when $d_X^{(i)} < \epsilon_{\mathrm{est}}$, $d_v^{(i)} < \epsilon_{\mathrm{est}}$, and $d_p^{(i)} < \epsilon_p$, and resets otherwise.

We also check the accepted, undamped site parameters. Let $\lambda_k = c_k^{-1}$ and $J_k = E_k/c_k$, and define

$$\begin{aligned} d_{\lambda,k}^{(i)} &= \frac{|\lambda_k^{(i)} - \lambda_k^{(i-1)}|}{\max\{\lambda_k^{(i)}, \lambda_k^{(i-1)}\}}, \\ d_{J,k}^{(i)} &= \frac{\left\| J_k^{(i)} - J_k^{(i-1)} \right\|_F}{\max\{1, \left\| J_k^{(i)} \right\|_F, \left\| J_k^{(i-1)} \right\|_F\}}, \\ d_{\mathrm{site}}^{(i)} &= \max_k\{d_{\lambda,k}^{(i)}, d_{J,k}^{(i)}\}, \qquad d_w^{(i)} = \max_k |p_k^{(i)} - w_k^{(i-1)}|. \end{aligned} \tag{61}$$

The site-stability counter increases only if the current site update is accepted, $d_{\mathrm{site}}^{(i)} < \epsilon_{\mathrm{site}}$, and $d_w^{(i)} < \epsilon_w$; it resets otherwise. After this check, update

$$w_k^{(i)} = (1-\zeta)w_k^{(i-1)} + \zeta p_k^{(i)}, \qquad 0 < \zeta \leq 1. \tag{62}$$

This history is used only for stopping; the receiver reports $p_k^{(i)}$ and retains the fixed prior ρ in (43).

Iteration stops after at least $I_{\min}$ iterations when both stability counters reach L_s and the hard activity vector has remained unchanged for L_d consecutive iterates. The two stability counters reset independently. If the iteration limit $I_{\max}$ is reached first, the final iterate is returned with that status. The damping weights, tolerances, and window lengths are given in the simulation setup.

C. Computational Complexity Analysis

We analyze the costs of phase estimation, noise estimation, and the subsequent MP iterations. Let G_{c} be the number of path matrices on the coarse delay–Doppler grid, $B \geq Q$ the number of phase bins, and S the number of coordinate sweeps. With dense path-matrix evaluation, precomputing these path matrices and their pilot responses costs $O(G_{\mathrm{c}}N^3 + KG_{\mathrm{c}}QN^2)$. Each user search forms correlations at cost $O(G_{\mathrm{c}}QNM)$ and evaluates phase bins by fast Fourier transforms (FFTs) at cost $O(G_{\mathrm{c}}MB\log B)$. An initial search orders the users, followed by S sweeps.

Let E_ϕ count all path-matrix evaluations for alias ranking and local refinement across the block, including derivative evaluations up to a constant factor. Evaluating a path matrix, computing its pilot response, and fitting the gain cost $O(N^3 + QN^2 + QNM)$. For $K \leq QN$, each joint gain refit costs $O(QNK^2 + QNKM)$. The resulting complexity of phase estimation is

$$\begin{aligned} C_{\mathrm{phase}} = O\big(&G_{\mathrm{c}}N^3 + KG_{\mathrm{c}}QN^2 \\ &+ (S+1)KG_{\mathrm{c}}M(QN + B\log B) \\ &+ E_\phi(N^3 + QN^2 + QNM) + S(QNK^2 + QNKM)\big). \end{aligned} \tag{63}$$

The local iteration budget bounds optimizer iterations; the actual number of objective evaluations also depends on line searches. Noise estimation costs $O(QNM^2 + M^3)$, and the thin SVD of $\widehat{\mathbf{A}}$ costs $O(KNQ^2)$ for $Q \leq KN$.

In each MP iteration, the dense linear transforms on MN measurement vectors cost $O(QKMN^2)$. The factors in (33) reuse the initial SVD.

Let C_θ, C_ψ, and C_ϕ denote the costs of one terminal-wise forward pass through the mean, error, and activity networks. Within the training variance range, each iteration evaluates the networks once per terminal and reuses F_θ

in the classifier. Taking into account the spatial covariance products and small-matrix operations, the computational complexity per iteration is given by

$$C_{\text{iter}} = O\big(QKMN^2 + KM^2N^2 + M^3 + K(C_\theta + C_\psi + C_\phi)\big)\,. \tag{64}$$

The networks are not evaluated outside the training variance range. For I MP iterations, the complete online cost is

$$O\big(C_{\text{phase}} + QNM^2 + M^3 + KNQ^2 + IC_{\text{iter}}\big)\,. \tag{65}$$

Excluding trained parameters and network activations, MP requires storage of $O(QKN + KMN^2 + KM^2)$ entries. The phase estimator additionally caches $O(G_{\text{c}}N^2 + KG_{\text{c}}QN)$ entries for the coarse path matrices and pilot responses, with $O(G_{\text{c}}BM)$ temporary FFT storage.

Appendix A
Second-Order Training Objective

The second equality in (53) follows from $F_{\theta,\mathbb{R}} - \mathbf{h}_{\mathbb{R}} = \sqrt{v_{\mathbb{R}}}\mathbf{b}$ and $2Dv_{\mathbb{R}} = Dv$.

For the fixed mean estimator F_θ, minimizing over scalar functions gives the population target

$$\tau_\theta(R, v) = \frac{1}{D}\mathbb{E}[\|F_\theta(R, v) - H\|_F^2 \mid R, v, \alpha = 1]. \tag{A.1}$$

Indeed, with $U = \|F_\theta - H\|_F^2/(Dv)$, conditioning (53) on $(R, v, \alpha = 1)$ gives $(a_\psi - \mathbb{E}[U \mid R, v, \alpha = 1])^2 + \mathrm{Var}(U \mid R, v, \alpha = 1)$. Moreover, expanding the squared error about the conditional mean gives

$$\tau_\theta(R, v) = \tau_{\text{C}}(R, v) + \frac{\|F_\theta(R, v) - m(R, v)\|_F^2}{D}, \tag{A.2}$$

where $\tau_{\text{C}} = D^{-1}\mathbb{E}[\|H - m\|_F^2 \mid R, v, \alpha = 1]$. The scalar output in (52) therefore learns the conditional mean-squared error of the deployed mean estimator. This target reduces to the posterior covariance trace per complex coordinate when $F_\theta = m$.

To derive the average in (40), fit Gaussian distributions with fixed means F_k and covariance $\tau\mathbf{I}_D$ for every terminal. Giving equal weight to each terminal's channel distribution conditioned on being active yields the expected negative log density, up to a constant,

$$\mathcal{J}(\tau) = KD\log\tau + \frac{D}{\tau}\sum_{k=1}^{K}\tau_\theta(R_k, v). \tag{A.3}$$

Its derivative is $KD(\tau - K^{-1}\sum_k \tau_\theta(R_k, v))/\tau^2$. The minimizer under $\tau \geq \tau_{\min}$ is the average conditional error clipped below by $\tau_{\min}$. Equation (40) approximates this fit with the learned error estimates. Equal weighting concerns each terminal's distribution conditioned on being active; activity probabilities enter later through the mixture in (44).

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